

Activation-Error Propagation for H-SVDQuant: From Local Reconstruction to Cumulative Block Reconstruction

Companion note to `hsvdquant.tex`

Abstract

This note studies the sequential calibration objective

$$\min \left\| \mathbf{X}_l \mathbf{W}_l - \hat{\mathbf{Y}}_l \right\|_F^2, \quad \hat{\mathbf{Y}}_l = \mathcal{Q}_A(\widetilde{\mathbf{X}}_l) \mathcal{Q}_W(\mathbf{R}_l) + \widehat{\mathbf{X}}_l \mathbf{L}_{1,l} \mathbf{L}_{2,l},$$

where $\mathbf{X}_l$ is the full-precision reference activation and $\widehat{\mathbf{X}}_l$ is the activation produced by already-quantized upstream operators. The main result is that the deterministic part of this apparently new problem is exactly the old H-SVDQuant problem with two substitutions: use the propagated Hessian $\widehat{\mathbf{H}}_l = \widehat{\mathbf{X}}_l^\top \widehat{\mathbf{X}}_l$, and replace $\mathbf{W}_l$ by a projected cumulative target $\mathbf{W}_l^\star = \widehat{\mathbf{H}}_l^\dagger \widehat{\mathbf{X}}_l^\top \mathbf{X}_l \mathbf{W}_l$. A non-removable projection error is added to the objective, but it is constant for the current layer. Consequently the low-rank refit, Hessian deflation, GPTQ feedback, smoothing gauge, and the joint $F_W + F_A$ code update all retain their form.

1 Reference and propagated paths

Suppress the block/linear index l . Let

$$\mathbf{X}, \widehat{\mathbf{X}} \in \mathbb{R}^{m \times c}, \quad \mathbf{W} \in \mathbb{R}^{c \times n}, \quad \mathbf{Y} = \mathbf{X} \mathbf{W}.$$

$\mathbf{X}$ is captured from the untouched full-precision path; $\widehat{\mathbf{X}}$ is produced by the partially quantized path. Let $\mathbf{D} = \text{diag}(d_1, \dots, d_c)$ and $\widetilde{\mathbf{X}} = \widehat{\mathbf{X}} \mathbf{D}^{-1}$. To match the objective in the question, write the full-precision branch in unsmoothed coordinates as $\mathbf{B} = \mathbf{L}_1 \mathbf{L}_2$ and the quantized residual in smoothed coordinates as $\widehat{\mathbf{R}} = \mathcal{Q}_W(\mathbf{R})$. Then

$$\hat{\mathbf{Y}} = \mathcal{Q}_A(\widetilde{\mathbf{X}}) \widehat{\mathbf{R}} + \widehat{\mathbf{X}} \mathbf{B}. \tag{1}$$

The implementation stores the branch in smoothed coordinates, $\widetilde{\mathbf{B}} = \mathbf{D} \mathbf{B} = \widetilde{\mathbf{L}}_1 \mathbf{L}_2$, so its runtime expression $(\widehat{\mathbf{X}} \mathbf{D}^{-1}) \widetilde{\mathbf{L}}_1 \mathbf{L}_2$ is identical to $\widehat{\mathbf{X}} \mathbf{B}$.

The old local objective uses $\widehat{\mathbf{X}} \mathbf{W}$ as its target. It therefore teaches the current layer to preserve its original weights under the new input, but does not ask it to compensate for upstream error. The cumulative objective instead keeps the full-precision target $\mathbf{X} \mathbf{W}$:

$$\mathcal{J}_{\text{cum}} = \left\| \mathbf{X} \mathbf{W} - \mathcal{Q}_A(\widetilde{\mathbf{X}} \mathbf{D}^{-1}) \widehat{\mathbf{R}} - \widehat{\mathbf{X}} \mathbf{B} \right\|_F^2. \tag{2}$$

2 Exact deterministic reduction

First turn off activation quantization and define the effective unsmoothed matrix

$$\mathbf{A} := \mathbf{B} + \mathbf{D}^{-1}\widehat{\mathbf{R}}, \quad \widehat{\mathbf{H}} := \widehat{\mathbf{X}}^\top \widehat{\mathbf{X}}, \quad \mathbf{C} := \widehat{\mathbf{X}}^\top \mathbf{X}, \quad \mathbf{G} := \mathbf{X}^\top \mathbf{X}.$$

The deterministic cumulative risk is

$$\begin{aligned} \mathcal{J}_0(\mathbf{A}) &= \|\mathbf{X}\mathbf{W} - \widehat{\mathbf{X}}\mathbf{A}\|_F^2 \\ &= \text{tr}(\mathbf{W}^\top \mathbf{G} \mathbf{W}) - 2\text{tr}(\mathbf{A}^\top \mathbf{C} \mathbf{W}) + \text{tr}(\mathbf{A}^\top \widehat{\mathbf{H}} \mathbf{A}). \end{aligned} \quad (3)$$

Proposition 1 (Projected cumulative target). *If $\widehat{\mathbf{H}} \succ 0$, define*

$$\boxed{\mathbf{A}^\star = \widehat{\mathbf{H}}^{-1} \mathbf{C} \mathbf{W} = \mathbf{W} + \widehat{\mathbf{H}}^{-1} \widehat{\mathbf{X}}^\top (\mathbf{X} - \widehat{\mathbf{X}}) \mathbf{W}.} \quad (4)$$

Then

$$\mathcal{J}_0(\mathbf{A}) = \underbrace{\|\widehat{\mathbf{H}}^{1/2}(\mathbf{A} - \mathbf{A}^\star)\|_F^2}_{\text{optimizable by the current layer}} + \underbrace{\text{tr}[\mathbf{W}^\top (\mathbf{G} - \mathbf{C}^\top \widehat{\mathbf{H}}^{-1} \mathbf{C}) \mathbf{W}]}_{\mathcal{J}_{\text{irr}}: \text{irreducible at this layer}}. \quad (5)$$

For singular $\widehat{\mathbf{H}}$, the same statement holds with the Moore–Penrose inverse and $\mathcal{J}_{\text{irr}} = \|(I - \widehat{\mathbf{X}}\widehat{\mathbf{X}}^\dagger)\mathbf{X}\mathbf{W}\|_F^2$.

Proof. The normal equation of (3) is $\widehat{\mathbf{H}}\mathbf{A} = \mathbf{C}\mathbf{W}$, which gives (4). Substitute $\mathbf{A} = \mathbf{A}^\star + (\mathbf{A} - \mathbf{A}^\star)$ in (3); the cross term vanishes by the normal equation, yielding (5). The singular case is ordinary least squares projected onto $\text{range}(\widehat{\mathbf{X}})$. $\square$

Equation (4) is the cumulative-error corrector. If $\widehat{\mathbf{X}} = \mathbf{X} + \delta\mathbf{X}$ and the perturbation is small, then to first order

$$\mathbf{A}^\star - \mathbf{W} \simeq -(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \delta\mathbf{X} \mathbf{W}. \quad (6)$$

Thus the current weight update is the least-squares inverse image of the upstream activation error. Error orthogonal to $\text{range}(\widehat{\mathbf{X}})$ cannot be repaired by any current linear map and belongs to $\mathcal{J}_{\text{irr}}$.

2.1 Reduction to the original smoothed problem

Set

$$\widetilde{\mathbf{W}}^\star := \mathbf{D}\mathbf{A}^\star, \quad \widetilde{\mathbf{B}} := \mathbf{D}\mathbf{B}, \quad \widetilde{\mathbf{H}} := \mathbf{D}^{-1}\widehat{\mathbf{H}}\mathbf{D}^{-1}. \quad (7)$$

Since $\mathbf{A} = \mathbf{D}^{-1}(\widetilde{\mathbf{B}} + \widehat{\mathbf{R}})$,

$$\mathcal{J}_0 = \mathcal{J}_{\text{irr}} + \left\| \widetilde{\mathbf{H}}^{1/2}(\widetilde{\mathbf{W}}^\star - \widetilde{\mathbf{B}} - \widehat{\mathbf{R}}) \right\|_F^2. \quad (8)$$

This is exactly the original H-SVDQuant deterministic objective, with

$$\boxed{\widetilde{\mathbf{H}} = \mathbf{D}^{-1}\mathbf{X}^\top \mathbf{X} \mathbf{D}^{-1} \longrightarrow \widetilde{\mathbf{H}} = \mathbf{D}^{-1}\widehat{\mathbf{X}}^\top \widehat{\mathbf{X}} \mathbf{D}^{-1}, \quad \widetilde{\mathbf{W}} = \mathbf{D}\mathbf{W} \longrightarrow \widetilde{\mathbf{W}}^\star = \mathbf{D}\widehat{\mathbf{H}}^\dagger \widehat{\mathbf{X}}^\top \mathbf{X} \mathbf{W}.} \quad (9)$$

The additive constant $\mathcal{J}_{\text{irr}}$ changes reported absolute error but cannot change the minimizer for the current layer.

Corollary 1 (Recovery of the local objective). *If $\widehat{\mathbf{X}} = \mathbf{X}$, then $\mathbf{A}^\star = \mathbf{W}$ and $\mathcal{J}_{\text{irr}} = 0$. All formulas reduce to the original local H-SVDQuant derivation.*

3 Activation quantization and the two-channel risk

Let

$$\mathbf{E}_A := \mathcal{Q}_A(\widetilde{\mathbf{X}}) - \widetilde{\mathbf{X}}, \quad \mathbb{E}[\mathbf{E}_A] = 0, \quad \Sigma_A := \mathbb{E}[\mathbf{E}_A^\top \mathbf{E}_A].$$

Using (1), the exact output error is

$$\mathbf{X}\mathbf{W} - \widehat{\mathbf{Y}} = (\mathbf{X}\mathbf{W} - \widehat{\mathbf{X}}\mathbf{A}^\star) + \widetilde{\mathbf{X}}(\widetilde{\mathbf{W}}^\star - \widetilde{\mathbf{B}} - \widehat{\mathbf{R}}) - \mathbf{E}_A \widehat{\mathbf{R}}.$$

The first two deterministic terms are orthogonal by the normal equation. Under the usual zero-mean/independence model for activation quantization noise, the noise cross terms also vanish. Therefore

$$\mathbb{E} \mathcal{J}_{\text{cum}} = \mathcal{J}_{\text{irr}} + \underbrace{\text{tr}(\Delta^\top \widetilde{\mathbf{H}} \Delta)}_{F_W^{\text{cum}}} + \underbrace{\text{tr}(\widehat{\mathbf{R}}^\top \Sigma_A \widehat{\mathbf{R}})}_{F_A^{\text{cum}}}, \quad \Delta = \widetilde{\mathbf{W}}^\star - \widetilde{\mathbf{B}} - \widehat{\mathbf{R}}. \quad (10)$$

The decomposition keeps its old form, but both channels are now conditioned on the propagated path: F_W uses $\widetilde{\mathbf{H}}$ and the shifted target $\widetilde{\mathbf{W}}^\star$, while F_A uses the noise covariance of $\mathcal{Q}_A(\widetilde{\mathbf{X}}\mathbf{D}^{-1})$. This matters because compensation can increase the norm or dynamic range of the residual, reducing deterministic accumulated error while increasing activation error. The existing balance coefficient λ is therefore more, not less, important:

$$\mathcal{J}_\lambda = F_W^{\text{cum}} + \lambda F_A^{\text{cum}}, \quad (11)$$

with $\mathcal{J}_{\text{irr}}$ omitted during optimization because it is constant.

4 What changes in each H-SVDQuant block

4.1 Low-rank refit and Hessian deflation

Write $\widetilde{\mathbf{B}} = \mathbf{L}_1 \mathbf{L}_2$. For fixed $\mathbf{L}_1$ and $\widehat{\mathbf{R}}$, let $\mathbf{M} = \mathbf{L}_1^\top \widetilde{\mathbf{H}} \mathbf{L}_1$. The exact post-code update remains

$$\mathbf{L}_2^\star = \mathbf{M}^{-1} \mathbf{L}_1^\top \widetilde{\mathbf{H}} (\widetilde{\mathbf{W}}^\star - \widehat{\mathbf{R}}). \quad (12)$$

Eliminating $\mathbf{L}_2$ gives the same deflated metric

$$\widetilde{\mathbf{H}}_\perp = \widetilde{\mathbf{H}} - \widetilde{\mathbf{H}} \mathbf{L}_1 (\mathbf{L}_1^\top \widetilde{\mathbf{H}} \mathbf{L}_1)^{-1} \mathbf{L}_1^\top \widetilde{\mathbf{H}}. \quad (13)$$

Hence weighted SVD, the $\widetilde{\mathbf{H}}$ -orthogonal gauge, and rank- r branch updates are unchanged algebraically. They are simply applied to $(\widetilde{\mathbf{H}}, \widetilde{\mathbf{W}}^\star)$ rather than $(\widetilde{\mathbf{H}}, \widetilde{\mathbf{W}})$.

4.2 Joint code target

After eliminating $\mathbf{L}_2$, the idealized code block is

$$\min_{\widehat{\mathbf{R}} \in \mathcal{Q}_W} \left\| \widetilde{\mathbf{W}}^\star - \widehat{\mathbf{R}} \right\|_{\widetilde{\mathbf{H}}_\perp}^2 + \lambda \|\widehat{\mathbf{R}}\|_{\Sigma_A}^2. \quad (14)$$

Completing the square gives

$$\mathbf{M}_Q = \widetilde{\mathbf{H}}_\perp + \lambda \Sigma_A, \quad \boxed{\mathbf{T}_Q = \mathbf{M}_Q^{-1} \widetilde{\mathbf{H}}_\perp \widetilde{\mathbf{W}}^\star}, \quad (15)$$

so GPTQ/GANQ should quantize $\mathbf{T}_Q$ under metric $\mathbf{M}_Q$. This is the same joint $F_W + F_A$ pseudo-target as before, except that the cumulative target $\widetilde{\mathbf{W}}^\star$ replaces the original weight. Positive activation noise also regularizes the rank-deficient null space of $\widetilde{\mathbf{H}}_\perp$.

4.3 Smoothing

For fixed $\widehat{\mathbf{X}}$ and $\mathbf{A}^\star$, changing $\mathbf{D}$ is still a gauge transform:

$$\widetilde{\widehat{\mathbf{X}}} = \widehat{\mathbf{X}}\mathbf{D}^{-1}, \quad \widetilde{\mathbf{W}^\star} = \mathbf{D}\mathbf{A}^\star, \quad \widetilde{\widehat{\mathbf{X}}}\widetilde{\mathbf{W}^\star} = \widehat{\mathbf{X}}\mathbf{A}^\star.$$

Thus $\mathbf{D}$ cannot change the unquantized deterministic optimum or $\mathcal{J}_{\text{irr}}$. It still changes weight group ranges and the covariance $\boldsymbol{\Sigma}_A$ through the quantizer, so the existing $F_W + \lambda F_A$ smoothing optimization remains valid after replacing the residual relative to $\mathbf{W}$ by the residual relative to $\mathbf{A}^\star$.

5 Stable finite-sample estimator

Computing a pseudoinverse from a small cache is unstable. Let the propagated streaming Hessian and paired-cache cross-error be

$$\widehat{\mathbf{H}} = \frac{1}{N} \sum_{t=1}^N \widehat{\mathbf{x}}_t \widehat{\mathbf{x}}_t^\top, \quad \widehat{\mathbf{K}} = \frac{1}{M} \sum_{t \in \mathcal{C}} \widehat{\mathbf{x}}_t (\mathbf{x}_t - \widehat{\mathbf{x}}_t)^\top.$$

The implemented estimator is

$$\boxed{\widehat{\mathbf{A}}_\varepsilon^\star = \mathbf{W} + (\widehat{\mathbf{H}} + \varepsilon \mathbf{I})^{-1} \widehat{\mathbf{K}} \mathbf{W}.} \quad (16)$$

It retains the full-token curvature used by GPTQ while estimating only the upstream mismatch on a bounded, paired reservoir. The reference and propagated reservoirs use identical random priorities, so their rows remain aligned.

The damping has a statistical meaning beyond numerical Cholesky stability: it shrinks compensation in activation directions that are weakly observed under $\widehat{\mathbf{X}}$. Without it, a small singular value can turn a modest upstream mismatch into a very large correction, inflate quantization ranges, and increase F_A .

6 Sequential block and linear calibration

For decoder block l , maintain two paths:

1. Run the untouched block on reference input $\mathbf{X}_l$ once. Cache the reference input of every linear and save the full-precision block output $\mathbf{X}_{l+1}$.
2. Run the working block on propagated input $\widehat{\mathbf{X}}_l$. Process true dependency groups in order: attention input projections (q, k, v) , attention output projection, MLP input projections (gate, up), and MLP output projection.
3. Before each group, collect its propagated Hessians/caches using all already-quantized predecessor groups. For each linear, construct (16), run H-SVDQuant, and replace the module immediately.
4. Run the fully quantized block on $\widehat{\mathbf{X}}_l$ to obtain $\widehat{\mathbf{X}}_{l+1}$.

Parallel projections are not artificially ordered: $q/k/v$ share the same input, as do gate/up, so they are calibrated independently from the same cache. Only genuine graph dependencies are propagated. This is the block analogue of GPTQ error feedback, while (16) is its activation-space, cross-layer counterpart.

7 Predictions, risks, and ablations

Block input	Linear objective	Interpretation
reference	local	no inter-block feedback control
quantized	local	strategy (a): propagated calibration only
reference	cumulative	intra-block accumulated-error compensation
quantized	cumulative	strategies (a)+(b), full sequential compensation

The theory gives several falsifiable predictions.

1. The cumulative mode should reduce the measured paired target error $\|\mathbf{X}\mathbf{W} - \hat{\mathbf{Y}}\|_F^2$, but need not reduce the local error $\|\hat{\mathbf{X}}\mathbf{W} - \hat{\mathbf{Y}}\|_F^2$.
2. Gains should grow with depth while correction norms remain controlled; no gain at the first parallel group is expected because $\hat{\mathbf{X}}_1 = \mathbf{X}_1$.
3. Very large correction-norm ratios diagnose ill-conditioned compensation and predict worse weight/activation quantization. Increasing damping or clipping the correction is then preferable to increasing rank.
4. Because compensation can move mass into the low-precision residual, the best λ under the cumulative objective may be larger than the best λ under the local objective.
5. Report both total paired MSE and its decomposition into upstream MSE, projected (irreducible) MSE, quantized state MSE, and correction-norm ratio. Perplexity alone cannot distinguish compensation from overfit.

8 Summary of theoretical changes

Object	Local reconstruction	Cumulative reconstruction
Target output	$\hat{\mathbf{X}}\mathbf{W}$	$\mathbf{X}\mathbf{W}$
Effective weight target	$\mathbf{W}$	$\mathbf{A}^* = \hat{\mathbf{H}}^\dagger \hat{\mathbf{X}}^\top \mathbf{X}\mathbf{W}$
Curvature	input Hessian of the local cache	$\hat{\mathbf{H}} = \hat{\mathbf{X}}^\top \hat{\mathbf{X}}$
Unavoidable term	zero	$\mathcal{J}_{\text{irr}}$
Smoothed target	$\mathbf{D}\mathbf{W}$	$\mathbf{D}\mathbf{A}^*$
Low-rank/ $\mathbf{L}_2$ updates	original formulas	unchanged, with target substitution
GPTQ metric	Hessian-deflated metric	unchanged, built from $\hat{\mathbf{H}}$
F_A covariance	noise of $\mathcal{Q}_A(\mathbf{X}\mathbf{D}^{-1})$	noise of $\mathcal{Q}_A(\hat{\mathbf{X}}\mathbf{D}^{-1})$
Joint code target	$\mathbf{M}_Q^{-1} \mathbf{H}_\perp \mathbf{D}\mathbf{W}$	$\mathbf{M}_Q^{-1} \mathbf{H}_\perp \mathbf{D}\mathbf{A}^*$

The central message is concise: cumulative activation-error correction does not require a new quantizer or a new low-rank solver. It requires a paired reference path, a propagated path, and the projected target (4). After that substitution, the existing H-SVDQuant machinery applies verbatim, with an added irreducible projection error that must be included when reporting absolute reconstruction quality.